(KTXFI1EBNF) Physics I. Lecture

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Current Updates and Issues

- Course materials (lecture and practice) are now available on Moodle, YouTube, and GitHub.
- Homework No. 1 has been released (Deadline: March 22, 2026, midnight). Supplementary material is provided to assist you.
- Please email your assignment to `facsko.gabor@uni-obuda.hu` as a **handwritten, scanned PDF**.
- Proposed midterm exam: **May 11, 2026**. Retake: **May 18, 2026**.

Where are we? I

1. **Mechanics I.:** Newton's laws, equation of motion (review). Momentum. General concept of mechanical work, work theorem. Forms of mechanical energy.
2. **Mechanics II.:** Weight force, gravitational force, gravitational field. Work of the gravitational force. Conservative force field. Dissipative force field. Mechanical power.
3. **Mechanics of Rigid Bodies I.** Concept of a rigid body. Forces acting on a rigid body. Motion of a rigid body. Concept of the centre of mass. Location of the centre of mass. Torque. Angular momentum. Energy of a rotating rigid body, angular momentum of a rotating rigid body, Steiner's theorem.
4. **Mechanics of Rigid Bodies II.** Principal theorems of rigid body motion; conservation laws. Collisions.
5. **Oscillations I.** Dynamic condition of harmonic oscillatory motion. Differential equation of harmonic undamped oscillatory motion.

Where are we? II

6. **Oscillations II.** Superposition of oscillations. Decomposition of oscillations into components. Damped oscillations. Forced oscillations, resonance. Coupled oscillations.
7. **Wave phenomena:** Reflection of waves, refraction of waves, diffraction of waves. Huygens–Fresnel principle. Interference of waves. Polarization of waves.
8. **Elements of acoustics:** Basics of acoustics. Fundamental concepts. Doppler effect. Sound intensity, loudness according to Weber–Fechner's law.
9. **Thermodynamics I:** Temperature scales, thermal expansion, thermometers. State variables, ideal gas, state changes, ideal gas equation of state, Boyle's law, Gay-Lussac's laws, universal (combined) gas law, heat, specific heat, molar heat, heat capacity.
10. **Thermodynamics II:** *Expansion work, internal energy, the first law of thermodynamics, forms of the first law in different state changes.*

Where are we? III

11. **Thermodynamics III:** Cyclic processes, Carnot cycle, Carnot's theorem, reduced heat, entropy, the second law of thermodynamics.
12. **Thermodynamics IV:** Kinetic theory of gases, molecular thermodynamics. Distributions, distribution functions. Elementary statistics.
13. **Electrostatics I.:** Electric charge: Fundamental electrical phenomena, electric charge and matter, Coulomb's law (force between two point charges, unit of electric charge, permittivity of vacuum).
14. **Electrostatics II.:** The electric field: The electrostatic field (electric field strength, electric field lines). Determination of electric field strength. Gauss's law (flux of the electric field). Electric field of a point charge.

Where are we? IV

15. **Electrostatics III.:** Electrostatic potential: Work of the electric field (work done in an electric field). Electric potential (electrostatic potential). Electric potential difference, voltage. Electric potential of charge systems at small and large distances. Electrostatic energy.
16. **Elements of physical or wave optics.** Branches of optics. Light interference, light diffraction, diffraction through a slit, optical grating, light polarization. Light as an electromagnetic wave: Poynting vector. Optical instruments.

Basics – types of thermodynamic systems I

- **Isolated system:** The thermodynamic system is isolated if there is no heat exchange or particle exchange between the gas and the surroundings.

  Symbol:

- **Closed system** A thermodynamic system is called closed if there is no particle exchange between the gas system and its surroundings.
- Heat and energy exchange between the gas and its surroundings are allowed.

  Symbol:

Basics – types of thermodynamic systems II

- **Open system:** A thermodynamic system is considered open if particle exchange and energy exchange are possible between the gas system and its surroundings.

  Symbol:

Volume work I

- Elemental work for small volume change:

$$W_i = -p_i \, dV_i$$

- Total work performed on the gas:

$$W = -\sum_{i=1}^{n} p_i \, dV_i$$

- In the limit of infinitesimal increments:

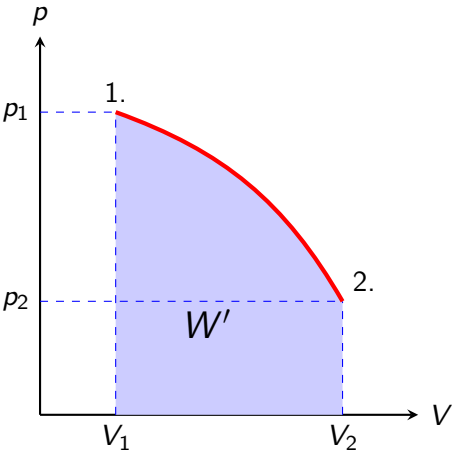
$$W = -\int_{V_1}^{V_2} p \, dV$$

- Work done by the gas:

$$W_g = -W = \int_{V_1}^{V_2} p \, dV$$

Volume work II

- Volume work on the p-V diagram
  - The numerical value of the volume work equals the area enclosed by the process curve in the p-V diagram.



Internal energy of ideal gas

The internal energy of ideal gases (U) originates from

- Kinetic energy of particles ($E_{kinetic}$)
- Potential energy of particles ($E_{potential}$)

Therefore,

$$U = E_{kinetic} + E_{potential}$$

Symbol: U
Unit: Joule (J)

The first law of thermodynamics I

- **Conservation of energy**
  - Macroscopic form:

$$\Delta U = Q + W$$

  - The energy of a thermodynamic system (U) is altered by the heat (Q) transferred to the system and the work (W) done on the system. In other words, the change in the system's internal energy equals the sum of the heat transferred to the system and the work done on the system.
  - This stament expresses the law of conservation of energy in thermodynamics.
  - Microscopic form:

$$dU = \delta Q + \delta W$$

  - d: the type of change (differential) is independent of the path (state function)
  - δ: the type of change (differentials) depends on the path (path function)

The first law of thermodynamics II

- **Isochoric ($V =$ const) process**
  - Volume does not change, hence

$$\begin{aligned} V &= \text{const} \\ dV &= 0 \end{aligned}$$

    where V is volume
  - Therefore,

$$dW = -p\,dV = 0,$$

    where W is work, p is pressure.
  - The first law of thermodynamics in this case:

$$dU = \delta Q + \delta W = \delta Q + 0 = \delta Q$$

The first law of thermodynamics III

- After integration:

$$dU = \delta Q \rightarrow U = Q$$

- Internal energy:

$$\begin{aligned} U &= c_V m T \\ \Delta U &= c_v m \Delta T, \end{aligned}$$

where m is mass, ΔT is temperature change, c_v is specific heat coefficient at constant volume.

The first law of thermodynamics IV

- **Isobaric ($p =$ const) process**
  - Pressure (p) does not change
    $$p = \text{const}$$
  - Elementary heat (Q, δQ):
    $$\delta Q = c_p m \Delta T \rightarrow Q = c_p m \Delta T,$$
    where m is mass, ΔT is temperature change, c_p is specific heat coefficient at constant pressure.
  - The volume work (W) done on the gas by the surroundings:
    $$W = -p\, dV,$$
    where p is pressure, dV is volume change.

The first law of thermodynamics V

- **Definition (Enthalpy, H)**: Enthalpy is a state function, which is defined as the follows:

$$H = U + pV,$$

where U is energy, p is pressure, V is volume, furthermore $[H] = J$.

  - From the first law of thermodynamics:

$$\begin{aligned} dU &= \delta Q + \delta W \\ dU - \delta W &= \delta Q \\ dU - (p \cdot dV) &= \delta Q \end{aligned}$$

  - We suppose that p=const, hence we can reorder for given heat the equation:

$$\begin{aligned} dU + (p \cdot dV) &= \delta Q \\ d\left(U + pV\right) &= \delta Q \\ dH &= \delta Q \quad \text{differential form} \end{aligned}$$

The first law of thermodynamics VI

- Finally,

$$dU = c_p \cdot m \cdot dT \rightarrow H = c_p \cdot m \cdot T.$$

- Enthalpy is useful because, at constant pressure, the heat added to the system is exactly equal to the change in enthalpy. This is why it is used in chemistry and engineering to describe most reactions (those occurring in open vessels at atmospheric pressure).

The first law of thermodynamics VII

- **Isothermal ($T =$ conts) process**
  - Temperature does not change
    $$T = \text{const}$$
  - Internal energy of ideal gas:
    $$dU = 0 \rightarrow \Delta U = 0$$
  - From the first law of thermodynamics:
    $$\begin{aligned} dU &= \delta Q + \delta W = 0 \\ \delta Q &= -\delta W = p \cdot dV \end{aligned}$$
  - The volumetric work of the gas done by the surroundings on the gas:
    $$W = -\frac{m}{M} RT \ln\left(\frac{V_{later}}{V_{earlier}}\right) = -\frac{m}{M} RT \ln\left(\frac{V_2}{V_1}\right)$$
  - Heat:
    $$Q = -W = \frac{m}{M} RT \ln\left(\frac{V_{later}}{V_{earlier}}\right) = \frac{m}{M} RT \ln\left(\frac{V_2}{V_1}\right)$$

The first law of thermodynamics VIII

- **Definition (Adiabatic Process):** An adiabatic process is a thermodynamics process where there is no heat exchange between the gas and its surroundings ($Q = 0$).
  - Internal energy of ideal gas: $dU = \delta Q + \delta W$, or $dU = 0 + \delta W = \delta W$.
  - For macroscopic quantities:

$$\Delta U = W = c_V m \Delta T$$

  - **Definition (Specific heat ratio):**

$$\kappa = \frac{c_p}{c_V} = \frac{C_p}{C_V}$$

The first law of thermodynamics IX

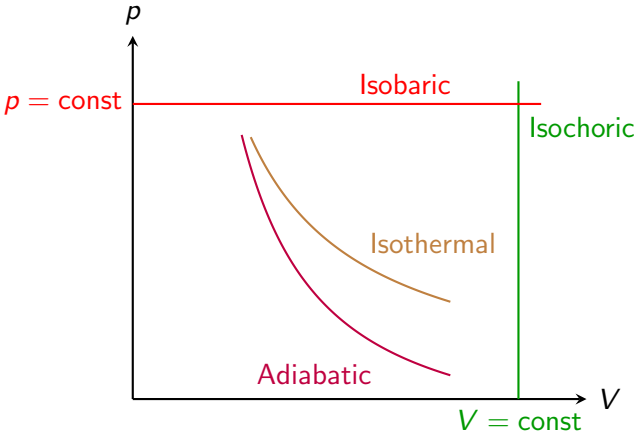
- **Poisson formulas:** The equations below describe an **adiabatic process** for an ideal gas.

$$
\begin{aligned}
p \cdot V^{\kappa} &= \text{const} \\
T \cdot V^{\kappa-1} &= \text{const} \\
T \cdot p^{\frac{1-\kappa}{\kappa}} &= \text{const},
\end{aligned}
$$

  where p is pressure, T is absolute temperature (K), V is volume, and κ (kappa) is heat capacity ratio (c_p/c_v).
- Examples:
  - **Adiabatic Compression:** If volume (V) decreases, the temperature (T) must increase (e.g., a bicycle pump getting warm).
  - **Adiabatic Expansion:** If volume (V) increases, the temperature (T) drops (e.g., the cooling effect of an aerosol spray).

The first law of thermodynamics X

- Curves of processes on the p-V state diagram



The End

Thank you for your attention!

Acknowledgements Thank you for the course materials for Szilárd ZSÓKA and Zoltán VARGA.